Paul Kiernan

Medium humanoid (engineer), chaotic good

Class & Level Staff Engineer 11 / CTO 3 • **Race** Human (Variant) • **Background** Sage (ECE) • **Alignment** Chaotic Good

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QUEST LOG

Sabbatical — The Long Rest

Nov 2025 – Present

- Taking my first proper rest in fifteen years. Family first, art second, and a slow-burn obsession with what happens to search infrastructure when the query interface starts talking back.
- Happy to trade notes on retrieval, Kafka war stories, or agentic developer tools. Say hi.

Phaselaw — The Startup Forge

Jul 2023 – Nov 2025

Chief Technology Officer

Oct 2024 – Nov 2025

- Architected an AI-powered document review and redaction platform entrusted with highly sensitive legal and personal data—10M+ documents securely redacted for the NHS, Labour (UK), and the Raspberry Pi Foundation
- Built and led the engineering organization; owned hiring, engineering culture, and technical due diligence with investors and partners
- Drove reliability and security practices—CI/CD, automated testing, multi-region infrastructure-as-code—through SOC 2 Type II certification

Founding Backend Engineer

Jul 2023 – Oct 2024

- Designed and built the core backend platform from zero—data models, API layer, auth, and service architecture deployed across multiple global regions
- Made foundational technology choices (language, framework, database, cloud provider), favoring boring, proven technology that let a tiny team ship and operate a global platform

Reddit Inc. — The Front Page Campaign

Jul 2018 – Jul 2023

Senior Staff Software Engineer

2021 – Jul 2023

- Built the technology that brought Reddit real-time comment indexing, a capability the company had lacked for a decade after post search: architected the Kafka infrastructure and consumer DAG that makes 7.5M+ new comments/day searchable within 60 seconds of posting
- The same indexing and retrieval systems later served all of Reddit's LLM applications

Staff Software Engineer — Data Engineering & Search Infrastructure

Nov 2019 – 2021

- Ran data engineering and search infrastructure: managed 3 engineers and mentored 3 more while staying hands-on. Owned end-to-end design of the Scala/Apache Flink pipeline keeping a multi-billion-document Solr/Lucene index fresh against millions of events per second
- Designed the Kafka platform that finally scaled for Reddit: multiple self-managed clusters with tuned partitioning and replication and a new consumer-processor pattern; a 5M+ messages/sec backbone adopted by the analytics, data warehousing, and subreddit notification teams
- Led search incident response and materially improved Reddit Search uptime

Senior Software Engineer II — Data Infrastructure

Jul 2018 – Nov 2019

- Replaced poll- and cron-based batch jobs with event-driven analytics and Kafka-to-BigQuery warehouse ingestion pipelines, cutting downstream latency from hours to seconds

Chartbeat Inc. — The Real-Time Chronicles

Aug 2014 – Jul 2018

Head of Backend Engineering

Oct 2016 – Jul 2018

- Forged *MrRoboto*, a chain-replicated C++ key-value store on LMDB handling 500k+ writes/sec
- Forged *Miata*, a real-time streaming analytics framework using Kafka, ZeroMQ, and Python
- Led a party of 8 backend engineers across 4 product teams, with weekly 1-on-1s for each

Backend Team Lead

Oct 2015 – Oct 2016

- Led the R&D team prototyping experimental products; presented research to the Chartbeat Board

Backend Developer

Aug 2014 – Oct 2015

- Open-sourced *Parselmouth*, an object-oriented interface for abstracting away the ugly parts of ad server APIs

Moat Inc. (acq. Oracle) — The Crawler Dungeons

Jun 2012 – Aug 2014

Software Developer

Jun 2012 – Aug 2014

- Early team member behind Moat Pro, an enterprise ad-intelligence platform for online advertising
- Engineered a network of web crawlers detecting and indexing online ads across geographic regions

ABILITY SCORES

SYSTEMS	DATA	BACKEND
18 (+4)	20 (+5)	18 (+4)

INFRA	LEADER	SCALE
17 (+3)	16 (+3)	19 (+4)

PERSONALITY

Traits. 15-year veteran of large-scale distributed systems. Founding engineer to CTO at Phaselaw. Senior Staff engineer at Reddit (500M+ monthly users): ran data engineering and search infrastructure—a multi-billion-document Solr/Lucene index and a 5M+ messages/sec Kafka backbone—and built the real-time comment indexing technology whose retrieval systems later served Reddit's LLM products. Inventor on US Patent 10,068,250.

Ideals. Do the simple thing that works. Boring technology, deployed carefully, measured honestly.

Bonds. Drawn to small, high-trust teams where rigor serves a mission.

Flaws. Physically incapable of walking past a flaky test.

PROFICIENCIES

Languages
Python • Scala • C++ • C
Java • Clojure • Shell • SQL

Data & Infrastructure
Kafka • Flink • Solr • Lucene
BigQuery • ZeroMQ • HAProxy
AWS • Docker • Kubernetes
Terraform • Temporal • ArgoCD
GenAI/LLM pipelines

Familiar
JavaScript • CSS • PHP

BACKSTORY

Cornell University
BS in ECE, 2008–2012
McMullen Dean Scholar
Goldfarb Tradition Fellow
Slope Media VP of Prod.
Nanosat 6 Power Subsystem
Cornell FSAE

SENDING STONES

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